

Quantitative Savings Results (Path A)

Eco-Loop / TwinPilot · HirePro Honeywell · <https://github.com/alphasafal/Twincity-/tree/ecoloop-hackathon-final>

Authoritative Path A metrics (comfort-zero)

Metric	Baseline	Agent	% reduction
Total energy (kWh)	421.5057	415.9999	1.31%
HVAC energy (kWh)	13.8459	13.157	4.98%
Peak power (kW)	19.9325	19.64	1.47%
Comfort violation hours	0.0	0.0	0

Speak line: AI proposes. SafetyShield validates. EnergyPlus executes.

Live dashboard: <https://twinpilot.webyaar.in> (login `manager@twinpilot.demo`)

Claims / methodology excerpt

Final Claims – Eco-Loop Building Agents

Primary claim (Path A – authoritative)

Under identical EnergyPlus building, weather, occupancy and simulation conditions, Eco-Loop reduced simulated HVAC energy by **4.98%**, total energy by **1.31%** and peak demand by **1.47%**, while maintaining **zero** occupied comfort-violation hours and **zero** comfort degree-hours.

Metric	Baseline	Eco-Loop	Improvement
-----	-----	-----	-----
Total energy	421.51 kWh	416.00 kWh	1.31%
HVAC energy	13.85 kWh	13.16 kWh	4.98%
Peak power	19.93 kW	19.64 kW	1.47%
Comfort violations	0 h	0 h	–
Comfort degree-hours	0	0	–
Actions	–	48 approved / 0 rejected / 0 fallback	–

Evidence: `evidence/independent-metrics.json`,
`evidence/energyplus/comparison.json`, `evidence/experiment-fairness.md`

Multi-scenario Path A (fair comparisons)

Scenario	Total saving	HVAC saving	Peak reduction	Comfort
-----	-----	-----	-----	-----

```
| Normal summer | 1.31% | 4.98% | 1.47% | 0 h |
| Hot / peak day | 1.27% | 5.07% | 1.08% | 0 h |
| High occupancy | 1.34% | 5.18% | 1.45% | 0 h |
```

Evidence: `results/scenarios/consolidated\_comparison.json`

Path C – hybrid supervisory (LLM materially in the loop)

Ollama selects energy-conservation strategies via MCP tools; the deterministic optimiser computes setpoints; SafetyShield gates actuation; expected-vs-actual outcomes are logged for self-correction.

Representative hybrid run (**not** a replacement for Path A claims):

- Separate MCP client/server PIDs (`mcp\_pids\_differ: true`)
- Strategies selected: COMFORT_FIRST, ECO_MODE, UNOCCUPIED_SETBACK, PRE_COOL, HOLD_CURRENT_POLICY
- Actions: **48 approved / 0 rejected / 0 fallback**
- HVAC $\approx$ **3.68%** / total $\approx$ **0.99%** vs same baseline (Path A remains the authoritative **4.98% / 1.31% / 1.47%** claim)
- Comfort violations: **0 h**
- Note: Path C peak can vary with strategy mix; do not substitute Path C peak for Path A
- Self-correction: `evidence/hybrid/self\_correction.jsonl`
- Loop excerpt: `evidence/hybrid/five-step-loop-excerpt.jsonl`
- Script: `./scripts/run\_hybrid\_supervisory\_experiment.sh`

Supporting claims

```
| Claim | Evidence |
|---|---|
| MCP separate stdio process | `evidence/mcp-process-proof.md`, `evidence/mcp-
runtime-trace.jsonl` |
| Ollama structured proposals | `evidence/ollama-runtime-proof.md` |
| SafetyShield rejects 35°C | `evidence/safety-rejection.log` |
| Actuator write to Clg-SetP-Sch | `evidence/final-actuator-trace.csv` |
| Next EnergyPlus state returned | `evidence/final-next-state-proof.md` |
| Ollama failure → deterministic fallback | `evidence/ollama-fallback.log` |
| Dashboard real data integrity | `evidence/dashboard-real-data-proof.md` |
| Dashboard honest no-data | `evidence/dashboard-no-data-proof.md` |
| Clean-clone | `evidence/clean-clone-report.md` |
| Carbon is estimated | comparison `carbon_accounting` (0.417 kg/kWh) |
| Hybrid supervisory Path C | `results/hybrid/summary.json`, `JUDGE_QA.md` |
```

Non-claims

- Not a physical BMS deployment.
- Not universal savings across buildings/weather.
- Not presenting 12.20% HVAC as comfort-zero authoritative.
- Not claiming Path A 4.98% was produced solely by the LLM writing every setpoint.
- Not calling in-process handler dispatch “MCP transport”.